MILITARY SPECIFICATION

GRAPHITE FIBER RESIN IMPREGNATED TAPE AND SHEET, FOR HAND LAYUP

1. SCOPE

1.1 Scope. This specification covers multifilament graphite tow and yarn reinforced resin matrix tape and sheet of 3 types and 4 classes for making low pressure laminates for use up to 420°F.

1.2 Application. Primarily for fabricating high strength and high modulus composite parts for aerospace application.

1.3 Classification. The multifilament graphite resin impregnated tape and sheet shall be of the type and class as specified.

TYPE I. GENERAL PURPOSE (for use at temperatures to 200°F)
TYPE II. HEAT RESISTANT (for use at temperatures to 350°F)
TYPE III. HEAT RESISTANT (for use at temperatures to 420°F)

Class A - 55 x 10^6 psi modulus of elasticity nominal; 250 x 10^3 psi tensile strength minimal (filament properties)
Class B - 75 x 10^6 psi modulus of elasticity nominal; 270 x 10^3 psi tensile strength minimal (filament properties)
Class C - 300 x 10^3 psi tensile strength minimal; 25 x 10^6 psi modulus minimal (filament properties)
Class D - 400 x 10^3 psi tensile strength minimal; 28 x 10^6 psi modulus minimal (filament properties)

(For the purpose of this specification nominal is defined as $\pm$ 5 x 10^6 psi.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-A-51	Acetone, Technical
TT-M-261	Methyl Ethyl Ketone, Technical

Military

MIL-Y-83371	Yarns, Graphite, High Modulus, Continuous Filament

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels
FED-STD-406	Plastics; Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents of the issue shown form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 972	Specific Gravity and Density of Plastics by Displacement
D2734	Void Content of Reinforced Plastics

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 General. The material covered by this specification shall consist of graphite fibers collimated parallel and in-plane, pre-impregnated with a thermosetting resin. They shall be capable of being molded, using low pressure laminating methods (under 200 psi) to produce a cured molding having properties described in this specification.

3.1.1 Procurement. Pre-impregnated graphite material shall be supplied by the linear foot in the width ordered.

3.2 Materials. The component materials used shall be as specified in 3.2.1 and 3.2.2 and shall be of high quality, entirely suited for the purpose.

3.2.1 Impregnating resin. The resins for Types I, II, and III materials shall conform to the properties shown in table I of this specification. The resin shall be free of foreign materials, noncorrosive to metals and shall be capable of being molded, using low pressure laminating methods, to a fully thermoset state and meet the requirements of this specification.

3.2.2 Graphite yarn and tows. The graphite fiber used shall meet the requirements of MIL-Y-83371 YARNS, GRAPHITE, HIGH MODULUS, CONTINUOUS FILAMENT.

3.2.3 Reinforced uncured pre-impregnated material, class A-D. The uncured graphite/resin prepreg material shall consist of graphite fibers collimated parallel in-plane and impregnated with a suitable resin (see 3.2.1).

a. The graphite fibers shall be completely wetted by the resin.

b. There shall be no cured resin particles in the material except for resin used for splicing yarns.

c. The prepreg shall be free of all parting agents or any other foreign material.

d. The physical properties of the uncured prepreg material shall be in accordance with the requirements shown in table I.

3.2.3.1 Physical properties. The as-received, pre-impregnated material shall meet the requirements of table I.

3.2.3.2 Graphite fiber splices. Fiber (yarn only) splices in the prepreg material shall be kept minimal to assure adherence to and attainment of the mechanical properties shown in table III of this specification. Splicing shall conform to the requirements specified in MIL-Y-83371.

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Table I. Physical properties (Types I, II, and III)

Property	Requirement
Tack (tape to tool)	Shall adhere for 30 minutes
Flow	± 3% of that specified
Volatile Content	± 2% of that specified
Resin Content	± 2% by weight of that specified within a range of 25% to 45%

3.2.3.3 Prepreg width. Prepreg tapes or sheet shall be furnished in the width specified. The width shall not vary more than ± .02 inch per inch of ordered width.

3.2.3.4 Prepreg length. Unless otherwise specified, tape rolls shall be 300 to 500 feet in length. However, within any one delivery within an order or order if completed in one shipment, one roll can be less than 300 feet but should not be less than 100 feet in length. The length of sheet shall not be less than the net length specified on the purchase order.

3.2.3.5 Prepreg uniformity. Each batch of prepreg material shall be of uniform quality throughout. Any section of prepreg material which does not meet the requirements of this specification shall not be removed from a continuous length of material. Linear footage of rejected material shall be itemized on the inspection tag and attached to the roll of prepreg and shall not be included in the specified length of prepreg material.

3.2.3.6 Cured prepreg physical properties. After fabrication of a cured composite in accordance with 4.8, the laminate shall meet the requirements of table II.

Table II. Physical properties required for cured composites

Properties	Average * Test Value
Per ply thickness	Determine
Fiber Volume	Nominal 55%
Void Content	Determine in accordance with ASTM D972
Density	Determine in accordance with ASTM D2734.

*Based on three (3) test samples

3.2.4 Mechanical properties. The cured laminates shall meet the requirements of table III.

3.3 Storage life (Shelf life). The prepreg material shall meet the requirements of this specification after storage of up to 3 months at temperatures not exceeding 0°F.

3.4 Working life. The materials shall possess sufficient room temperature working life such that the requirements of table III are met after exposure to 65°-80°F temperature and a relative humidity not in excess of 70 percent for a continuous period of 14 days and cured with the manufacturer's recommended cure cycle identical to that recommended for "unaged" material.

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TABLE III. MECHANICAL PROPERTIES OF CURED LAMINATES
CLASS A

PROPERTY	-67°F ± 10°F	ROOM TEMPERATURE	200°F ± 10°F	350°F ± 10°F	420°F ± 10°F
TENSILE STRENGH, (psi)					
Longitudinal	65,000	100,000	98,000	100,000	100,000
Transverse	4,500	4,000	3,500	3,000	3,000
TENSILE MODULUS, (psi)					
Longitudinal	23.5×10^6	25.5×10^6	25.5×10^6	25.5×10^6	25.5×10^6
Transverse	1.0×10^6	1.0×10^6	1.0×10^6	0.8×10^6	0.8×10^6
COMPRESSIVE STRENGTH, (psi)					
Longitudinal	92,000	100,000	90,000	85,000	80,000
Transverse		29,000	24,000	20,000	18,000
COMPRESSIVE MODULUS, (psi)					
Longitudinal	23×10^6	25.5×10^6	25.5×10^6	25×10^6	25×10^6
Transverse		1.0×10^6	1.0×10^6	0.9×10^6	0.85×10^6
FLEXURAL STRENGTH, (psi)					
Longitudinal	90,000	100,000	98,000	92,000	90,000
FLEXURAL MODULUS, (psi)					
Longitudinal	23.5×10^6	23×10^6	23.5×10^6	22.8×10^6	20×10^6
SHORT BEAM SHEAR STRENGTH, (psi)	8,500	7,500	6,900	4,700	4,500

NOTES:
1. Specified values are the minimum average of 4 determinations. Individual minimums shall not be less than 90 percent of the value specified.
2. Property values are based on a test laminate with a fiber volume of 55 percent nominally. When required, normalize test values to a laminate fiber volume of 55 percent.
3. Elevated temperature tests will be conducted after not less than 30 minutes exposure to the specified temperature.
4. Requirements for the specified types: TYPE I R.T. & 200°F, TYPE II R.T. & 350°F, TYPE III R.T. & 420°F.

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TABLE III (Continued). MECHANICAL PROPERTIES OF CURED LAMINATES
CLASS B

PROPERTY	-67 ± 10°F	ROOM TEMPERATURE	200 ± 10°F	350 ± 10°F	420 ± 10°F
TENSILE STRENGTH, psi					
Longitudinal	75,000	115,000	112,000	105,000	100,000
Transverse	3,500	3,000	2,800	2,500	2,500
TENSILE MODULUS, psi					
Longitudinal	35×10^6	38×10^6	38×10^6	37×10^6	36×10^6
Transverse	0.86×10^6	0.86×10^6	0.86×10^6	0.8×10^6	0.8×10^6
COMPRESSIVE STRENGTH, psi					
Longitudinal	73,000	80,000	72,000	70,000	65,000
Transverse		25,000	20,000	18,000	16,000
COMPRESSIVE MODULUS, psi					
Longitudinal	34.5×10^6	36×10^6	36×10^6	34.5×10^6	33×10^6
Transverse		0.86×10^6	0.86×10^6	0.8×10^6	0.76×10^6
FLEXURAL STRENGTH, psi					
Longitudinal	91,000	100,000	96,000	90,000	86,000
FLEXURAL MODULUS, psi					
Longitudinal	37×10^6	37×10^6	37×10^6	36.5×10^6	36×10^6
SHORT BEAM SHEAR STRENGTH, psi	6,000	5,500	5,100	3,500	3,300

NOTES:
1. Specified values are the minimum average of 4 determinations. Individual minimums shall not be less than 90 percent of the value specified.
2. Property values are based on a test laminate with a fiber volume of 55 percent nominally. When required, normalize test values to a laminate fiber volume of 55 percent.
3. Elevated temperatures tests will be conducted after not less than 30 minutes exposure to the specified temperature.
4. Requirements for the specified types: TYPE I R.T. & 200°F, TYPE II R.T. & 300°F, TYPE III R.T. & 400°F.

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TABLE III. (Continued). MECHANICAL PROPERTIES OF CURED LAMINATES
CLASS C

PROPERTY	-67 ± 10°F	ROOM TEMPERATURE	200 ± 10°F	350 ± 10°F	420 ± 10°F
TENSILE STRENGTH, psi					
Longitudinal	140,000	195,000	190,000	180,000	180,000
Transverse	8,000	8,000	7,000	6,500	6,000
TENSILE MODULUS, psi					
Longitudinal	14×10^6	16×10^6	16×10^6	15.5×10^6	15×10^6
Transverse	1×10^6	1×10^6	1×10^6	0.8×10^6	0.75×10^6
COMPRESSION STRENGTH, psi					
Longitudinal	140,000	160,000	150,000	110,000	105,000
Transverse		28,500	27,000	22,500	19,500
COMPRESSION MODULUS, psi					
Longitudinal	13×10^6	15.5×10^6	15.5×10^6	15×10^6	15×10^6
Transverse		2×10^6	2×10^6	1.8×10^6	1.75×10^6
FLEXURAL STRENGTH, psi					
Longitudinal	165,000	195,000	190,000	140,000	120,000
FLEXURAL MODULUS, psi					
Longitudinal	15×10^6	15×10^6	15×10^6	15×10^6	14×10^6
SHORT BEAM SHEAR STRENGTH, psi	8,500	10,000	9,200	7,000	6,000

NOTES:
1. Specified values are the minimum average of 4 determinations. Individual minimums shall not be less than 90 percent of the value specified.
2. Property values are based on a test laminate with a fiber volume of 55 percent nominally. When required, normalize test values to a laminate fiber volume of 55 percent.
3. Elevated temperature tests will be conducted after not less than 30 minute exposure to the specified temperature.
4. Requirements for the specified types: TYPE I R.T. & 200°F, TYPE II R.T. & 350°F, TYPE III R.T. & 420°F.

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TABLE III. (Continued). MECHANICAL PROPERTIES OF CURED LAMINATES
CLASS D

PROPERTY	-67 ± 10°F	ROOM TEMPERATURE	200 ± 10°F	350 ± 10°F	420 ± 10°F
TENSILE STRENGTH, psi					
Longitudinal	100,000	140,000	140,000	130,000	130,000
Transverse	4,700	4,500	4,000	3,000	2,500
TENSILE MODULUS, psi					
Longitudinal	16.5×10^6	18.5×10^6	18.5×10^6	18×10^6	18×10^6
Transverse	1.2×10^6	1.2×10^6	1.1×10^6	1×10^6	0.9×10^6
COMPRESSIVE STRENGTH, psi					
Longitudinal	100,000	110,000	105,000	100,000	95,000
Transverse		30,000	28,000	23,000	20,000
COMPRESSIVE MODULUS, psi					
Longitudinal	10×10^6	13×10^6	13×10^6	12.5×10^6	12×10^6
Transverse		1.2×10^6	1.2×10^6	1.1×10^6	1×10^6
FLEXURAL STRENGTH, psi					
Longitudinal	150,000	160,000	160,000	95,000	90,000
FLEXURAL MODULUS, psi					
Longitudinal	16×10^6	16×10^6	16×10^6	15.5×10^6	15×10^6
SHORT BEAM SHEAR STRENGTH, psi	8,700	8,500	8,000	6,200	5,200

NOTES:
1. Specified values are the minimum average of 4 determinations. Individual minimums shall not be less than 90 percent of the value specified.
2. Property values are based on a test laminate with a fiber volume of 55 percent nominally. When required, normalize test values to a laminate fiber volume of 55 percent.
3. Elevated temperature tests will be conducted after not less than 30 minute exposure to the specified temperature.
4. Requirements for the specified types: TYPE I R.T. & 200°F, TYPE II R.T. & 350°F, TYPE III R.T. & 420°F.

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3.5 Workmanship. Prepreg material furnished to this specification shall be of quality workmanship and shall be free of all impurities and defects which could adversely affect its performance. Visible indications or incomplete impregnation shall be marked by inserts and shall be cause for rejection only if the total length of such areas exceeds 2 percent of the total tape length of the roll.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certification of conformance. Unless otherwise specified in the purchase order, the supplier shall furnish with each shipment, a certified report (in triplicate) signed by a duly authorized representative of the supplier, stating conformance to the requirements specified herein and listing the specific results of all the acceptance tests. This report shall also include this specification number, the purchase order number, and the batch number.

4.3 Subcontractor. When materials for subcontract fabrication are purchased directly by the subcontractor, the subcontractor shall be responsible for determining that the material meets all the requirements of this specification. With each part shipment the subcontractor shall submit a copy of the report as specified in 4.2.

4.4 Inspection records. The supplier's inspection records of examination and tests for conformance to the requirements of this specification shall be kept complete and available to the purchaser upon request.

4.5 Re-inspection. For each original nonconforming specimen result, three additional test specimens shall be used to retest each nonconforming result. Failure of any retest specimen to meet the specific requirement shall be cause for rejection of the product represented and no additional testing shall be permitted. Results of all tests shall be reported.

4.6 Inspection lot. A lot may consist of material from more than one batch, forming part of one purchase order and submitted for acceptance at one time. Under these circumstances, each batch shall be subjected to the herein specified acceptance tests. A batch shall be that quantity of material compounded, manufactured and delivered at one time from one batch of resin under the same fixed conditions.

4.7 Test conditions

4.7.1 Standard conditions. Normal temperature tests shall be conducted at 70° to 80°F (21° to 27°C). If the validity of property values determined within this temperature range is questionable, the tests shall be repeated at standard conditions (23° ± 1°C and 50 ± 5 percent relative humidity.

4.8 Test specimen preparation. Test panels from which test specimen will be prepared shall be fabricated in a manner defined by the prepreg materials supplier, consistent with standard aircraft processing procedures and compatible with the physical and mechanical requirements of this specification. Processing shall include post cure if required of the materials system selected to develop optimum physical and mechanical properties.

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4.8.1 Completed test specimens. Using diamond tipped cutters or carbide edged tools, test specimens shall be cut from test panel prepared as specified in 4.8.

4.9 Testing

4.9.1 Uncured prepreg physical test specimens

Test	Specimen No. and Size
Volatile content	2 ply, 1 gram, approx.
Resin content	2 ply, 1 gram, approx.

Test	Specimen No. and Size
Flow	2 plies, 2 inch square stacked 0° and 90° alternately
Tack	2 plies, 1 inch long by 3 width and thickness of prepreg.

4.9.2 Cured prepreg mechanical test specimens. Test specimens shall be prepared in accordance with 4.8 and 4.8.1 of this specification. The specimens shall contain a nominal 55 percent graphite fiber volume.

Test	Paragraph	Fiber Direction
Flexure, Longitudinal	4.11.3.1	0°
Flexure, Modulus	4.11.3.1	0°
Shear, Short Beam	4.11.3.2	0°
Tension, Longitudinal	4.11.3.3	0°
Tension, Modulus	4.11.3.3	0°
Tension, Transverse	4.11.3.3	90°
Compression, Longitudinal	4.11.3.4	0°
Compression, Modulus	4.11.3.4	0°
Compression, Transverse	4.11.3.4	90°
Compression, Modulus	4.11.3.4	90°

4.10 Classification of tests. The inspection and testing of the herein specified material shall be classified as follows:

a. Preproduction tests (see 4.11).

b. Acceptance tests (see 4.12).

4.11 Preproduction tests. Preproduction tests shall be performed by the supplier and shall comply with all requirements specified in section 3. The supplier shall submit a test report and samples of the material to the purchaser for evaluation.

4.11.1 Re-inspection. Once a product is approved, any change in formulation shall be submitted by the manufacturer in writing to the purchaser. The material shall then be subjected to preproduction tests.

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4.11.2 Physical tests, prepreg material. Physical tests shall be performed on the prepreg materials using specimen sizes and quantity required for test as specified in 4.9 and tested as described below.

4.11.2.1 Resin flow. Resin flow shall be determined in accordance with the procedures described below:

1. Material:

   a. Aluminum foil, one piece 6 by 12 inches.

   b. Suitable fiberglass bleeder cloth, four pieces 3 inches square.

   c. Suitable release cloth, tetrafluoroethylene (TGE) two pieces 3 inches square.

   d. Specimen, two pieces - 2 inches square.

2. Equipment:

   a. Platen Press, model SP-215, Pasadena Hydraulic, Inc., or equivalent.

   b. Precision Balance, Mettler, Type B5, or equivalent.

   c. Stop Watch, Model Number NE-13D, or equivalent, Northeast Instrument Company.

3. Specimen Preparation:

   a. Remove separator paper from one side of the sample material and sandwich the exposed sides together with their fibers oriented at 0° - 90°.

   b. Remove the separator paper from one side of the specimen sandwich and weigh the specimen to the nearest milligram (mg). Place the side of the specimen that still has release paper attached on the balance pan.

   c. Crease the sheet of aluminum foil in the center.

   d. Stack 2 pieces of the fiberglass bleeder cloth on one another and lay them on top of the aluminum foil aligning an edge with the center crease of the foil.

   e. Center 1 piece of the TFE release cloth on top of the fiberglass bleeder cloth.

   f. Center the exposed side of the specimen on top of the TFE release cloth and remove the remaining piece of release paper from the specimen. Weigh the release paper to the nearest mg.

   g. Center 1 piece of TFE release cloth on top of specimen and place 2 pieces of fiberglass bleeder cloth on top of the TFE release cloth.

   h. Fold the aluminum foil over the layup to form a 6 by 6-inch square.

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4. Procedure:

   a. Preheat the platen press to the temperature recommended by the resin manufacture.

   b. Place the layup on the press and immediately apply the pressure (recommended by the resin manufacture) to the specimen. Start the stop watch when the pressure is applied.

   c. When the layup has been in the press for 15 minutes ± 15 seconds, remove it from the press and allow to cool to room temperature.

   d. Remove any resin that has extruded from the body of the specimen and is clinging to the edges. Remove the specimen from the layup and weigh to the nearest mg.

5. Calculation:

$$\text{Resin flow, percent} = \frac{(W_1 - W_2) - W_3}{W_1 - W_2} \times 100$$

where W_1 = initial weight of specimen with one piece of release paper still attached, grams

W_2 = weight of release paper of W_1, grams

W_3 = final weight of specimen after pressing, grams,

Report the average of a minimum of 2 determinations.

4.11.2.2 Volatile content. Volatile content shall be determined in accordance with the procedures described below:

1. Material:
   a. Tray
   b. Film, parting agent
   c. Specimen, approximately 1 gram

2. Equipment:
   a. Oven, circulating air
   b. Desiccator
   c. Precision Balance

3. Procedure:

   Cut sufficient material to obtain a specimen approximately 1 gram and weigh to nearest milligram (mg) W_1. Suspend the specimen on a removable tray (area of the tray in contact with the specimen shall be covered with a suitable parting agent film) in a circulating air oven preheated to a temperature commensurate with the matrix system utilized. Heat the specimen for a period of 60 minutes ±5. Remove specimen from the tray and place in a desiccator. Cool specimen to room temperature inside the desiccator for at least 30 minutes and weigh to the nearest mg W_2.

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4. Calculation:

$$\text{Volatile Content, \% by weight} = \frac{W_1 - W_2}{W_1} \times 100$$

where, W_1 = Original weight specimen, mg

W_2 = Final weight of specimen, mg

4.11.2.3 Resin (Non-fiber) content. Resin (non-fiber) content shall be determined by either Procedure A (solvent extraction) or Procedure B (acid digestion) described below:

Procedure A. Solvent Extraction

1. Material:
   a. Glass beaker
   b. Solvent
   c. Specimen, two pieces - approximately 1 gram each

2. Equipment:
   a. Hot plate
   b. Desiccator
   c. Circulating air oven
   d. Precision Balance

3. Procedure:
   Cut sufficient material to obtain two specimens each weighing approximately 1 gram and weigh to the nearest milligram (mg) W_1. Place the specimens in separate containers and wash with a suitable boiling solvent for not less than 2 minutes. Time starts when the solvent starts to boil. Decant the solvent. The solvent used shall be selected on the basis of conditions of this test. Repeat this procedure three additional times. Place the specimens in a circulating air oven preheated to $325^{\circ} \pm 10^{\circ}$ F for a period of not less than one hour or until weight is constant. Place the specimens in a desiccator for ten minutes. Remove and weigh to the nearest mg W_2.

4. Calculation:
   Resin (non-fiber) Content, % by weight $= \frac{W_1 - W_2}{W_1} \times 100$

   where, W_1 = Original weight of specimen, mg
   W_2 = Final weight of specimen, mg

5. Reporting:
   Report both the individual results and arithmetic mean of the two determinations.

Procedure B. Acid digestion for resin systems which cannot be solvent extracted.

1. Reagent: Nitric acid, sulfuric acid, or sulfuric acid/hydrogen peroxide can be used. The digestion temperature should not exceed 140°F when nitric acid is used.

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2. Procedure: Two pieces, each approximately one gram in weight, shall be cut from the tape. Remove the separator film from the specimen and weigh in separate, suitably cleaned, tared filtering vessels. Dissolve and digest the resin material in the selected digestion media. Digestion should continue until fibers are completely separated as determined by visual examination. Filter and wash fibers. Oven dry, cool in desiccator, and weigh.

3. Reporting: Report both the individual results and arithmetic mean of the two determinations.

4.11.2.4 Tack, tape to tool

a. Cut 2 pieces of impregnated graphite material approximately 1.0 by 3.0 inch retaining the protective film until immediately before using the specimens. Fiber direction shall be parallel to the 1.0 inch dimension. For material less than 3.0 inch wide, butt sufficient pieces on the panel to produce a 3.0 inch wide specimen.

b. Remove the protective film from one side of one specimen and apply the material to the center of a clean piece of corrosion resistant steel sheet, type 302 or equivalent, with a commercial 2D finish, any thickness by approximately 4 by 8 inches. Apply light pressure with a squeegee or roller over the backing film. Remove the backing film and apply the second specimen to the first, in exactly the same manner, making sure the opposing faces of the material are not covered with protective backing film. The second layer of narrow material shall be so positioned that the butt joints do not coincide. Remove the protective film from the exposed surface of the material and maintain the test plate and the long dimension of the test specimen in a vertical position for not less than 30 minutes at 21.1° - 26.7°C (70° - 80°F) and 50-70 percent relative humidity.

c. Report results as pass or fail. If a specimen fails to adhere for the test period, record the lapsed time at failure.

4.11.3 Mechanical tests, cured composite laminate. Mechanical test shall be performed on cured laminates made from material supplied to this specification and cured as recommended by the manufacturer. Specimen sizes and quantities required for test are specified in 4.9.

4.11.3.1 Flexure, longitudinal (0°) strength and modulus

1. Specimen Preparation: The specimens, taken from the laminate prepared as specified in 4.8, shall be 0.500 inch ± 0.010 inch wide and 3.00 inch ±0.03 long. The thickness shall be that of the laminate and shall not vary by more than ± .005 inch. Specimen edges shall be ground to the required length and width dimensions with abrasive finer than 400 grit. The fiber direction of the specimen shall be parallel to the longitudinal axis of the specimen. Other specimen configurations are acceptable provided that the fiber direction is parallel to the longitudinal axis of the specimen, the thickness of the specimen is equal to or less than the width of the specimen, and the length of the specimen is sufficient to provide a span equal to 32 ± 2 times the thickness. Flexural test specimen shown in figure 1.

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2. Test Procedure: The specimen shall be loaded to failure at a crossheat speed of 0.050 inch ± 0.005 inch per minute in a testing machine with fixture and instrumentation as indicated in figure 2. The deflectometer shall be of the linear differential transformer type. Necessary adjustments shall be made so that the deflectometer pushrod is midway between the supports ± 0.03 inch. The span length shall be 32 ± 2 times the specimen thickness.

3. Calculation: Flexural strength and modulus shall be obtained by the formulas given below:

$$\text{Flexural strength, psi} = \frac{3PL}{4\,bt^2}$$

$$\text{Flexure Modulus of Elasticity, psi} = \frac{11}{64}\,\frac{L^3}{bt^3}\,\frac{(\Delta P)}{(\Delta \delta)}$$

where, P = Load at failure, lb
L = Span, in.
b = Width, in.
t = Thickness, in.
P = Increment of load, lb
ΔP = Increment of deflection, in.
$\Delta \delta$ = Slope of initial straight line portion of the load deflection curve, lb per in.

4. Reporting: Calculate the arithmetic mean of 4 determinations as the flexural strength and modulus of elasticity of the sample.

4.11.3.2 Short beam shear (apparent horizontal shear)

1. Specimen Preparation: The specimen, taken from the laminate prepared as specified in 4.8 shall be 0.250 inch ± 0.005 inch wide and 0.700 inch ± 0.010 inch long. Specimen edges shall be ground to the required length and width dimensions with abrasive finer than 400 grit. The fibers shall be parallel to a longitudinal axis. Short beam shear specimen shown in figure 3.

2. Test Procedure: A 3-point loading device as shown in figure 4 with loading nose and supports of diameters indicated shall be used. The specimen shall be loaded at the center of a span 4 ± 0.1 times the thickness of the specimen. The loading device shall provide for accurate centering and alignment. The centerlines of the cylindrical surfaces of the loading nose and the supports shall be parallel. The longitudinal axis of the specimens shall be perpendicular to the centerlines of the cylindrical surfaces within ± 0.017 rad (± 1 degree). The specimens shall be loaded to failure at a crosshead speed of 0.050 inch ± 0.005 inch per minute.

3. Calculation: The short beam shear strength shall be calculated as follows:

$$\text{Short Beam Shear Strength, psi} = \frac{3P}{4A}$$

where, P = Load at failure, lb

A = Specimen cross-sectional area, sq. in.

4. Reporting: Calculate the arithmetic mean of 4 determinations as the short beam shear strength of the sample. Report the individual test results and the arithmetic mean.

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4.11.3.3 Tension, strength and modulus

1. Specimen Preparation: The specimens taken from the laminate prepared as specified in 4.8 shall be straight sided coupons. Specimen edges shall be ground to the required width and length dimensions with abrasive finer than 400 grit.

2. Longitudinal Specimen (0°): Test specimens shall conform to figure 5, with adhesive bonded laminate tabs. Specimen fibers shall be parallel to the longitudinal axis.

3. Transverse Specimen (90°): Test specimens shall conform to figure 6. Specimen fibers shall be normal to the longitudinal axis. Tabs are not used.

4. Procedure: The specimen shall be loaded to failure at a 0.050 inch $\pm$ 0.005 inch per minute crosshead speed in a testing machine, using instrumentation as shown in figures 5 and 6. Test temperatures shall be as specified in table III. Specimens shall be tested after exposure for not less than 30 minutes at the test temperature.

5. Calculation: Tensile strength and modulus of elasticity shall be obtained by the formulas given below:

   Ultimate Tensile Strength, psi $= \frac{F_t}{A}$

   Tensile Modulus of Elasticity, psi $= \frac{F_t}{E}$

   where, F_t = Maximum tensile load, lb
   A = Specimen cross-sectional area, sq. in.
   E = Corresponding strain in inch per inch

6. Obtain the modulus of elasticity by extending the initial straightline portion of the load-deflection curve and graphically determining the ratio of stress to corresponding strain.

7. Reporting: Calculate the arithmetic mean of 4 determinations for each test as tensile strength and modulus of elasticity of the sample. Report both the individual test results and the arithmetic mean.

4.11.3.4 Compression, strength and modulus

1. Specimen Preparation: The specimens taken from the laminate prepared as specified in 4.8 shall be straight sided coupons. Specimens edges shall be ground to the required width and length dimensions with abrasive finer than 400 grit.

2. Longitudinal Specimens (0°): Test specimens shall conform to figure 7. Specimen fibers shall be parallel to the longitudinal axis.

3. Transverse Specimens (90°): Test specimens shall conform to figure 8. Specimen fibers shall be normal to the longitudinal axis.

4. Test Method: The test method used shall be reported. The method identified as "Celanese Compression Test" is preferred.

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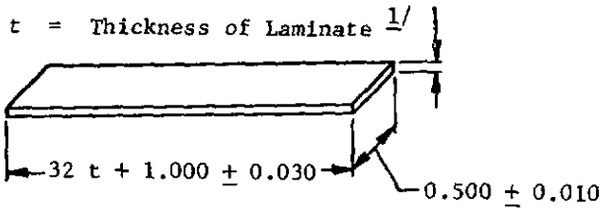


FIGURE 1. Flexural test specimen

1/ Specimen thickness shall not vary more than ± 0.005 inch

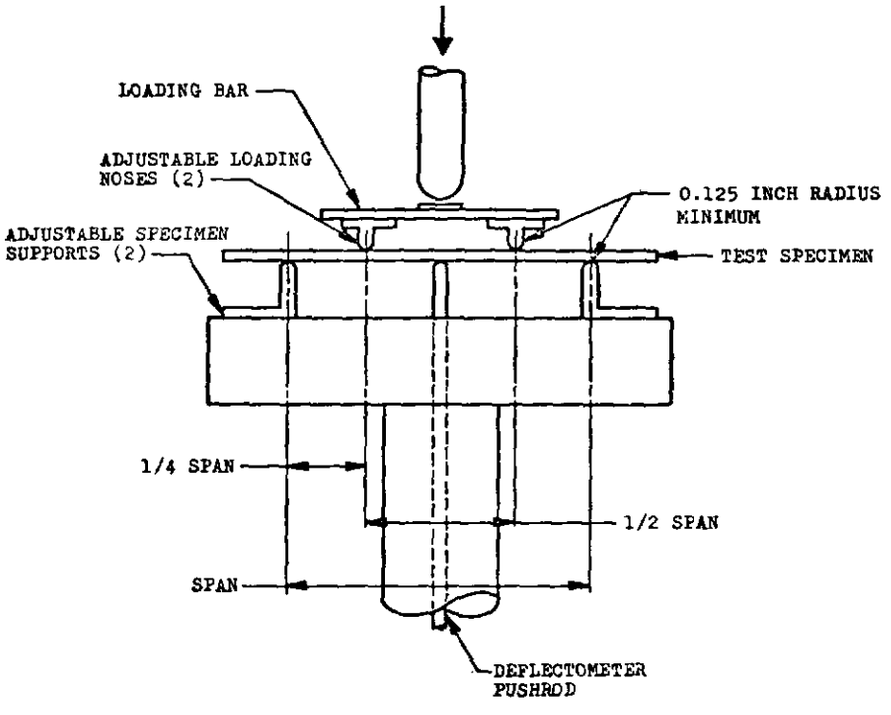


FIGURE 2. Flexural test fixture setup

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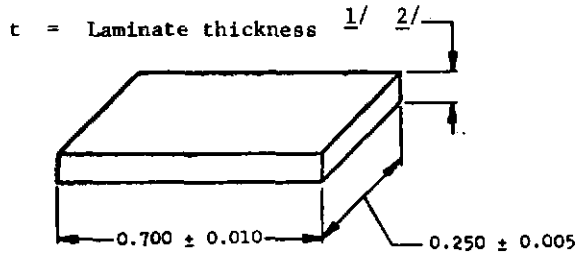


FIGURE 3. Short beam shear test specimen

1/ Specimen thickness shall not vary more than ± 0.005 inch.

2/ Dimensions are based on a 0.080 inch nominal thickness.

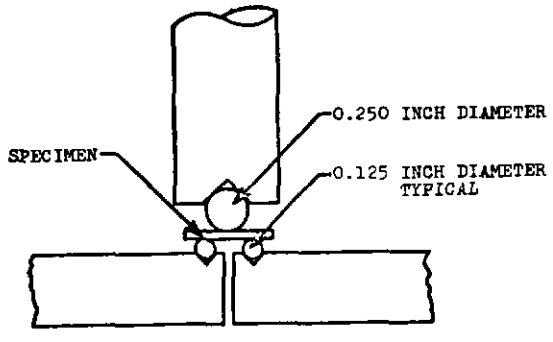


FIGURE 4. Short beam shear testing fixture

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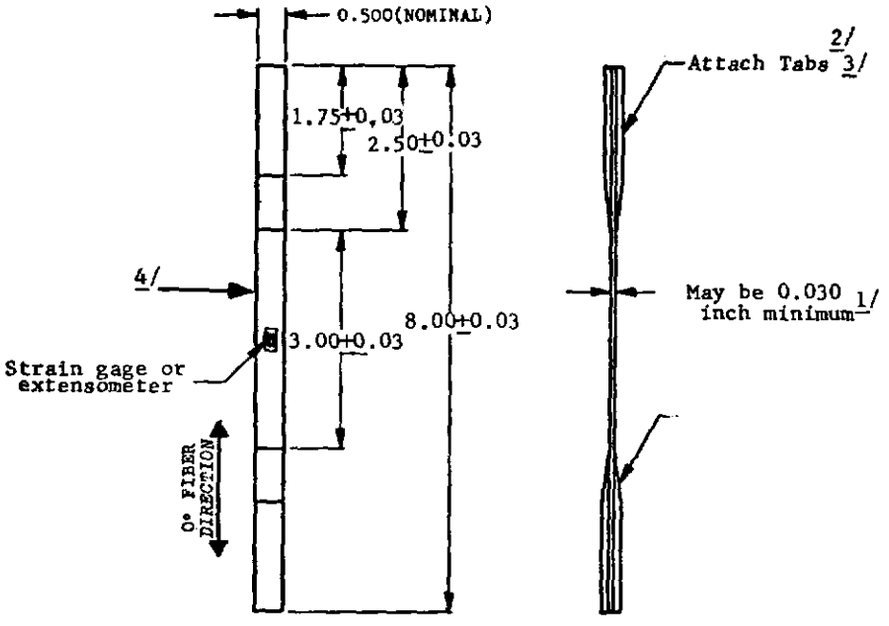


FIGURE 5. Longitudinal tensile test specimen

1/ Ply thickness based on 0.005 inch per ply.

2/ Tabs of laminated E glass fibers with epoxy resin suitable for 350°F ± 10, fibers oriented at right angles with specimen axis, 4 ply tabs on both sides of both ends regardless of specimen thickness.

3/ Machine chamfer to feather edge the end of each laminated tab towards the specimen center and attach with heat resistant epoxy adhesive prior to final machining of test specimen.

4/ Specimen edges shall be parallel to ± 0.002 inch. Polish edges after machining.

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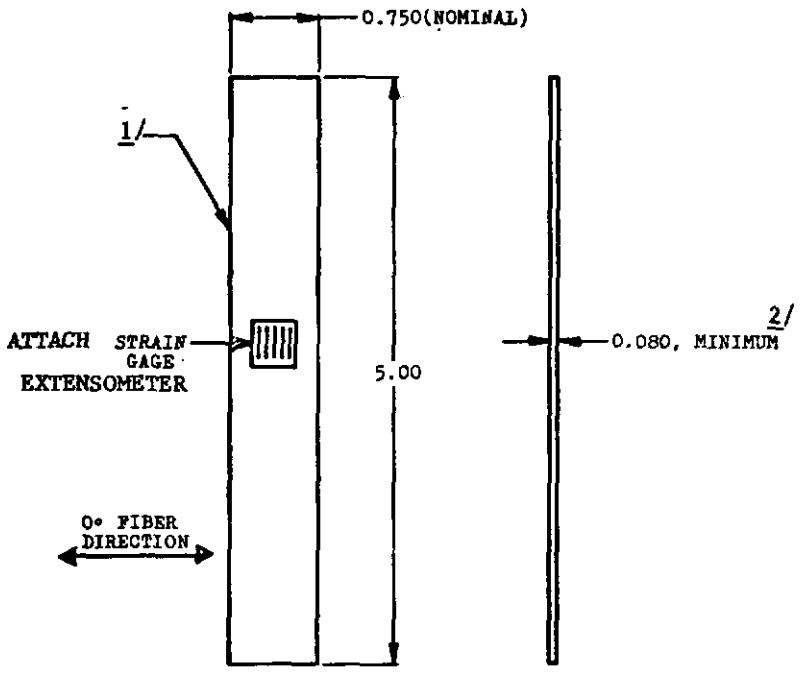


FIGURE 6. Transverse tensile test specimen

1/ Specimen longitudinal edges shall be parallel to 0.030 inch.

2/ Specimen thickness shall not vary more than ± 0.005 inch.

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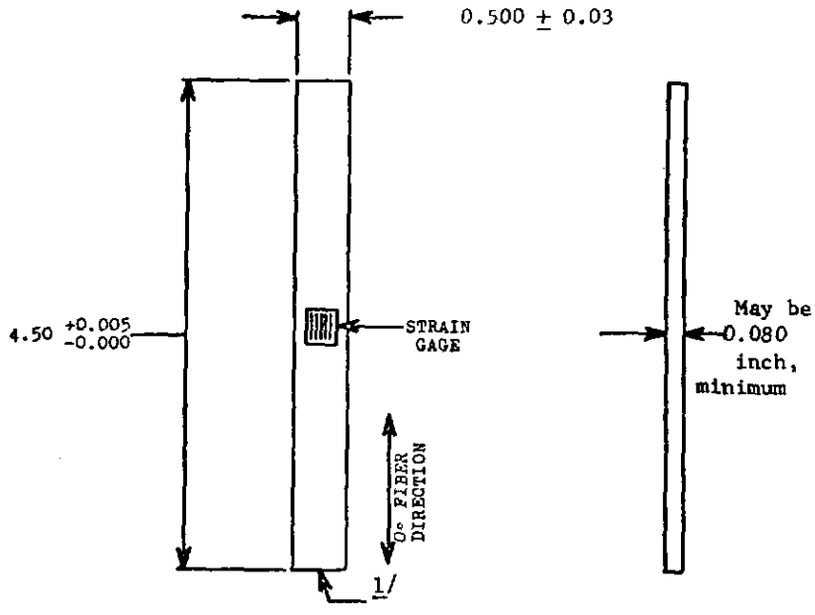


FIGURE 7. Longitudinal compression test specimen

1/ Top and bottom surfaces shall be parallel and flat to 0.005 inch.

2/ Specimen longitudinal edges shall be parallel to 0.003 inch.

3/ Specimen thickness shall not vary more than 0.005 inch from nominal.

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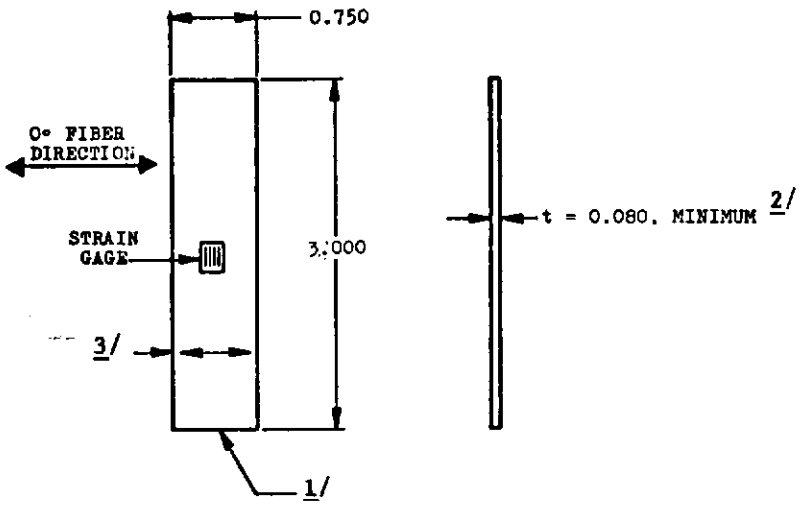


FIGURE 8. Transverse (90°) compression test specimen

1/ Top and bottom surfaces shall be flat and parallel to 0.005 inch.

2/ Specimen thickness shall not vary more than 0.005 inch from nominal.

3/ Specimen longitudinal edges shall be parallel to ± 0.003 inch.

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5. Reporting: Calculate the arithmetic mean of 4 determinations for each test as compressive strength and modulus of elasticity of sample. Report both the individual test results and the arithmetic mean.

4.11.3.5 Fiber volume

1. Preparation of specimens: Three pieces, each not less than 1.0 inch square, shall be cut from the laminate prepared as specified in 4.8, and weighed in separate cleaned and tared (W_1) thimbles to the nearest mg (W_2).

2. Acid Digestion: Place each thimble in a separate 150 ml beaker and add 100 ml of acid to the beaker. The beaker with the contents shall be heated at a suitable temperature until the resin has dissolved and digestion is complete and cooled to room temperature. Nitric acid, sulfuric acid or sulfuric acid/hydrogen peroxide can be used. The digestion temperature should not exceed 140°F, when nitric acid is used.

3. Filtration: Discard the acid used in the digestion without loss of residue and attach the thimble on a vacuum filter flask. Rinse the fibers with additional clean acid, followed by distilled water, and finally with acetone. The acetone shall remain clear, indicating complete removal of resin from the fibers.

4. Drying and weighing: Each thimble, with the contents, shall be heated in a circulating air oven at 121.1° ± 5.6°C (250° ±10°F) for 15 minutes ± 1, cooled in a desiccator to room temperature for at least 60 minutes and weighed immediately to the nearest mg (W_3).

5. Calculation: The laminate fiber volume shall be calculated as follows:

$$\text{Fiber Volume, \%} = \frac{(W_3 - W_1)}{(W_2 - W_1)} \cdot \frac{\rho_2}{\rho_1} \times 100$$

where, W_1 = Weight of extraction thimble, mg

W_2 = Weight of extraction thimble plus specimen, mg

W_3 = Weight of extraction thimble plus specimen after digestion, mg

ρ_1 = Density of graphite tow as specified in applicable detail specification of MIL-Y-83371

ρ_2 = Density of laminate determined by ASTM D792

6. Reporting: Calculate the arithmetic mean of 3 determinations as the fiber volume of the test laminate. Report the individual test results and the arithmetic mean.

4.11.3.6 Laminate density: Determine the laminate density in accordance with ASTM D2734.

4.11.4 Test results report: In addition to the requirements of 4.2, the vendor of the product shall include the identification of the resin system and the cure cycle used to fabricate the laminate.

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4.12 Acceptance tests. The following tests are classified as acceptance tests. A random sample from each production lot shall be selected for inspection. Inspections shall be made for conformance to requirements for Identification Marking, Packaging and Packing (5.1 and 5.2). The roll of prepreg shall then be allowed to reach standard temperature before breaking the seal. Open seal and test prepreg length in accordance with 3.2.3.4. Remove 5 square feet of prepreg. Reseal the package of prepreg and return to $0^{o}F$ storage. Identify the 5 square foot sample and proceed to inspect for the following requirements:

1. Prepreg width - (see 3.2.3.3).

2. Uniformity - (see 3.2.3.5).

3. Uncured prepreg physical properties - (see 3.2.3).

4. Cured prepreg flexural strength at 0° to filament direction - (see table III) at R.T. and appropriate elevated temperature as specified by Type.

4.13 Retest. If a sample fails to meet two or less requirements of this specification, a retest is permitted. A second failure shall be cause for rejection. The results of the original tests and the retest results shall be included in the test report.

4.14 Rejection. Each batch of material shall be rejected if it does not pass the acceptance tests subject to the retest provisions as specified in 4.13.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Prepreg material tape shall be rolled on a reel of not less than 8 inches in diameter. A non-adherent, non-porous, and non-transferable paper or separator of a contrasting color shall be used on one side of the material to prevent the layers of material from sticking to each other. Unless otherwise specified, each roll and each sheet of material shall be sealed in a suitable non-adherent material to prevent penetration of moisture or loss of impregnating resin solvent. An identification tag shall be placed within each bag prior to sealing. Sheet material may be shipped flat.

5.1.1 Identification marking. Each roll of prepreg shall be permanently marked with the following data:

SPECIAL INSTRUCTIONS (is required, see 5.2.1)

RESIN TREATED GRAPHITE REINFORCED PREPREG. LOW PRESSURE MOLDING

PURCHASER DESIGNATION NO.

DATE OF IMPREGNATION

RESIN CONTENT, PERCENT

VOLATILES, PERCENT

FLOW, PERCENT

WIDTH OF MATERIAL

LINEAR FEET

MANUFACTURER'S BATCH NO. AND DATE OF MANUFACTURE

MANUFACTURER'S DESIGNATION

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STORAGE TEMPERATURE, MAX: $0^{o}F$

SHELF LIFE: 3 months

INSPECTION RECORD AND COMMENTS: ITEMIZED DESCRIPTION OF

REJECTED MATERIAL INCLUDING LINEAR FOOTAGE OF SUCH REJECTS AND THEIR LOCATION.

5.2 Packing. Units packaged as specified in 5.1 shall be packed in exterior-type shipping containers in a manner that will allow solid carbon dioxide to be packed in sufficient quantities to maintain a material temperature of $0^{o}F$, minimum, during transit. Upon receipt, containers shall be opened and examined to ascertain that solid carbon dioxide remains therein, if used. Prepreg rolls shall be packed in a horizontal position and containers so marked so as to ensure horizontal positioning for shipment and later stored in an upright, vertical position. The shipping container shall be so constructed as to assure safe delivery and acceptance at their destination. Shipping containers shall comply with carrier regulations applicable to the mode of transportation.

5.2.1 Marking of shipment. Each shipping container shall be marked with the following information and in accordance with MIL-STD-129.

GRAPHITE FIBER RESIN IMPREGNATED TAPE AND SHEET, FOR HAND LAYUP

PURCHASE ORDER NO.

MANUFACTURER'S NAME, TRADEMARK OR SYMBOL

MANUFACTURER'S BATCH AND LOT NO.

NOMINAL WIDTH AND LENGTH PER ROLL

STORAGE TEMPERATURE, MAXIMUM $0^{o}F$

DATE OF MANUFACTURE

SHELF LIFE: 3 months

FRAGILE

6. NOTES

6.1 Intended use. The material procured in accordance with this specification is intended for use in airframe and aerospace primary structural components and in similar assemblies where high stiffness and strength-to-weight ratios are required.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.

b. Type and classes.

c. Width.

d. Length.

e. Quantity and size of containers required.

f. Packaging and packing (see 5.1 and 5.2).

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